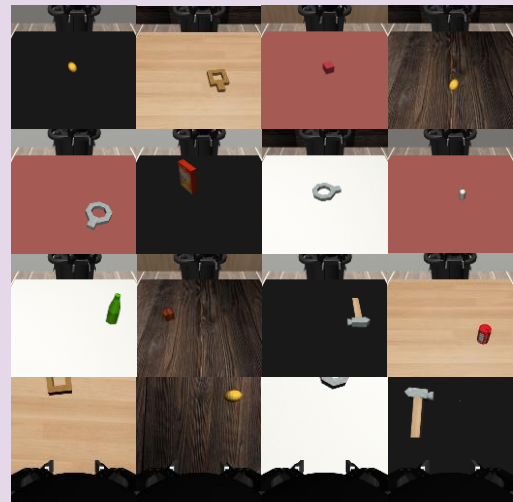




The experimental setup involves **11 objects**, **5 environments** and **4 background materials**, which ensures the diversity of the dataset. The experimental procedure is controlled by a unified **seed chain** for reproducibility.

The **joint angles** of the robotic arm and **dual-view images** (a front view and a wrist view) are obtained via onboard sensors.

Observation



Closed-Loop Restoration

State

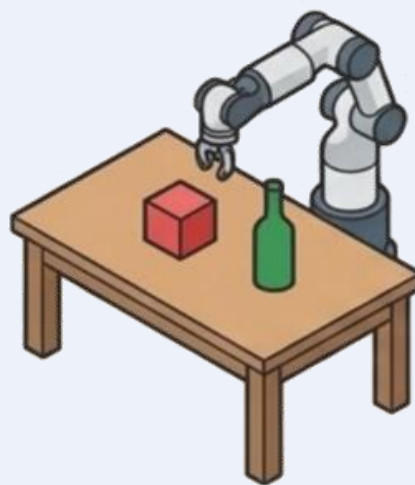
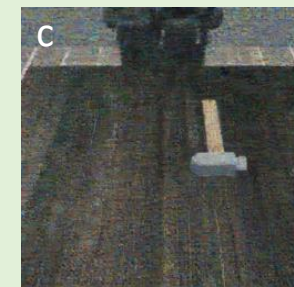
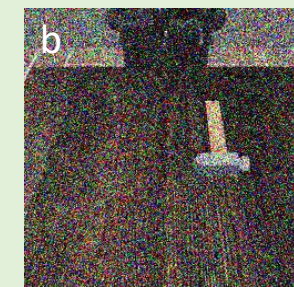


Image Pipeline

25
Distortions

13
Restoration



Add noise

Inference

Restoration

Inference

Experiments divide into three groups, **raw image(a)**, **distortion only(b)** and **restoration(c)**. These experiments were conducted within a **closed-loop benchmark framework**.

Action

Inference

Success: **True**
Steps: **50**

Success: **False**
Steps: **250**

3 VLAs(OpenVLA -oft, Pi-0FAST, Pi-0.5) generates **7-DoF actions** in real time from visual observations after pipeline and current state

The robotic arm executes actions and updates its next-step state in both **simulation** and the **real world**. Task performance is evaluated jointly by **success rate(SR)** and number of **steps** in the experiments.